

### Unit 3 — Numerical Methods (4 Qs)

**1. State the RK-4 formula for  $dy/dx = f(x,y)$  — Pure recall, foundation for every Unit 3 Part-C question.**  
**High.**

Answer:

$$y_{n+1} = y_n + (1/6) (k_1 + 2k_2 + 2k_3 + k_4)$$

Where:

- $k_1 = h f(x_n, y_n)$
- $k_2 = h f(x_n + h/2, y_n + k_1/2)$
- $k_3 = h f(x_n + h/2, y_n + k_2/2)$
- $k_4 = h f(x_n + h, y_n + k_3)$

**2. Write Milne's Predictor and Corrector formulas — Matches Question Bank 'state predictor-corrector' style Q. High.**

Answer:

Predictor Formula:

$$y_{n+1}^{(P)} = y_{n-3} + (4h/3) [2f_{n-2} - f_{n-1} + 2f_n]$$

Corrector Formula:

$$y_{n+1}^{(C)} = y_{n-1} + (h/3) [f_{n-1} + 4f_n + f_{n+1}^{(P)}]$$

**3. Write Euler's Modified Method iterative formula — Directly needed to attempt Euler's Modified numerals. Medium-High.**

Answer:

$$y_{n+1}^{(k+1)} = y_n + (h/2) [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(k)})]$$

**4. State the Adams-Bashforth predictor formula — Less repeated than Milne's/RK4 in PYQs but still asked.**  
**Medium.**

Answer:

$$y_{n+1}^{(P)} = y_n + (h/24) [55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}]$$

## Unit 4 — Transform Calculus (Laplace & Fourier) (6 Qs)

### 5. State and explain the Convolution Theorem — ☐ Very High — repeated across PYQ 2023/2024/2025

Answer:

If  $L^{-1}\{F(s)\} = f(t)$  and  $L^{-1}\{G(s)\} = g(t)$ , then:

$$L^{-1}\{F(s)G(s)\} = \int_0^t f(u) g(t-u) du = f * g$$

### 6. Find $L\{t^2 e^{-3t}\}$ — ☐ Very High — exact match PYQ June 2023

Answer:

We know that  $L\{t^2\} = 2! / s^3 = 2 / s^3$

By the First Shifting Theorem,  $L\{e^{at}f(t)\} = F(s-a)$ . Here,  $a = -3$ .

Therefore,  $L\{t^2 e^{-3t}\} = 2 / (s+3)^3$

### 7. Find $L^{-1}\{1 / (1+s)^3\}$ — ☐ Very High — exact match PYQ June 2023

Answer:

We know that  $L^{-1}\{1 / s^3\} = t^2 / 2! = t^2 / 2$

By the First Shifting Theorem in inverse form,  $L^{-1}\{F(s-a)\} = e^{at} f(t)$ . Here,  $a = -1$ .

Therefore,  $L^{-1}\{1 / (s+1)^3\} = (1/2) t^2 e^{-t}$

### 8. State First and Second Shifting Theorems — Foundational; used implicitly in almost every Part C Laplace question. High.

Answer:

First Shifting Theorem:

If  $L\{f(t)\} = F(s)$ , then  $L\{e^{at}f(t)\} = F(s-a)$

Second Shifting Theorem:

If  $L\{f(t)\} = F(s)$ , then  $L\{f(t-a)u(t-a)\} = e^{-as} F(s)$

**9. Find  $L\{t \cos(at)\}$  — High — PYQ June 2023**

Answer:

Using the property  $L\{t f(t)\} = -d/ds F(s)$ :

$$L\{\cos(at)\} = s / (s^2 + a^2)$$

$$L\{t \cos(at)\} = -d/ds [ s / (s^2 + a^2) ]$$

$$= - [ (s^2 + a^2)(1) - s(2s) ] / (s^2 + a^2)^2$$

$$= (s^2 - a^2) / (s^2 + a^2)^2$$

**10. Find the Fourier Sine Transform of  $f(x) = 1/x$  — Standard Dirichlet-integral type. Medium.**

Answer:

$$F_s\{1/x\} = \int_0^\infty f(x) \sin(sx) dx = \int_0^\infty (\sin(sx) / x) dx$$

Using the standard Dirichlet integral result:

$$F_s\{1/x\} = \pi/2$$

**Unit 5 — Concept of Probability & Statistics (5 Qs)****11. Find  $c$  for  $f(x) = cx^2$ ,  $0 < x < 1$  as a valid PDF — ☐ Very High — exact match PYQ Dec 2025 + Question Bank**

Answer:

For a valid PDF, the total area under the curve must be 1.

$$\int_0^1 cx^2 dx = 1$$

$$c [ x^3 / 3 ]_0^1 = 1$$

$$c (1/3 - 0) = 1 \Rightarrow c / 3 = 1 \Rightarrow c = 3$$

**12. State four properties of the Normal Distribution curve — High**

Answer:

- It is bell-shaped and perfectly symmetrical about its mean ( $\mu$ ).
- The Mean, Median, and Mode are all equal and located at the center of the distribution.
- The curve is asymptotic to the horizontal x-axis (it approaches but never touches it).
- The total area under the probability density curve is equal to 1.

**13. Write short note on Exponential Distribution — PDF and Mean — ☐ Very High**

Answer:

The Exponential Distribution is a continuous probability distribution often used to model waiting times or lifetimes.

Probability Density Function (PDF):  $f(x) = \lambda e^{-\lambda x}$  for  $x \geq 0$

Theoretical Mean (Expected Value): Mean =  $1 / \lambda$

**14. Poisson distribution: If  $P(X=1) = P(X=2)$ , find the mean  $\lambda$  — Medium-High — clean, fast numeric type**

Answer:

The Poisson probability formula is  $P(X=k) = (e^{-\lambda} \lambda^k) / k!$

Given  $P(X=1) = P(X=2)$ :

$$(e^{-\lambda} \lambda^1) / 1! = (e^{-\lambda} \lambda^2) / 2!$$

$$\lambda = \lambda^2 / 2 \quad \square \quad \lambda^2 - 2\lambda = 0$$

Since  $\lambda > 0$ , we have  $\lambda = 2$ . Thus, the mean is 2.

**15. Binomial distribution: Mean = 4, Variance = 3, find n and p — Medium-High — same family as the 7-mark  $9P(X=4)=P(X=2)$  type, good 2-mark variant**

Answer:

For a Binomial distribution  $B(n, p)$ :

$$\text{Mean} = np = 4$$

$$\text{Variance} = npq = 3$$

$$\text{We know that } q = \text{Variance} / \text{Mean} = 3 / 4 = 0.75$$

$$\text{Since } p + q = 1, p = 1 - 0.75 = 0.25$$

$$\text{Substitute p into Mean: } n(0.25) = 4 \quad \square \quad n = 16$$